

CS & IT ENGINEERING

COMPILER DESIGN

PARSING

Lecture-04



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Recap of Previous Lecture



Topic

?? LR(0) Parsing

LR(0) items DFA

LR(0) Grammar detection

LR(0) Parsing table



Topics to be Covered



Topic

??

SLR(1) Parsing

Topic

??

SLR(1) Grammar

Topic

??

First set
Follow set



Topic : First Set

= First set of Non terminal is set of terminals which appears in First position on grammar R.H.S.

1. First abc = {a}
2. $A \rightarrow \underline{aX} \mid \underline{bY}$
First (A) = {a, b}
3. $A \rightarrow \underline{B}x$ $B \rightarrow \epsilon$
First(A) = First(B)
4. $A \rightarrow \cancel{B}X$ $B \rightarrow \epsilon$
First(A) = {First(B) - ϵ } $\cup$ {First(X)}

$$\textcircled{2} \quad S \rightarrow A \underline{B}$$

$$F(S) = F(A) \\ \{a, \epsilon\}$$

$$A \rightarrow a | \epsilon$$

$$B \rightarrow d$$

$$F(S) = \{a, d\}$$

$$F(A) = \{a, \epsilon\}$$

$$F(B) = \{d\}$$

$$F(S) = \underline{F(A)} \cup F(B)$$

$$\{a, \epsilon\} \quad \{b, \epsilon\}$$

$$\textcircled{3} \left\{ \begin{array}{l} S \rightarrow AB \\ A \rightarrow a | \epsilon \\ B \rightarrow b | \epsilon \end{array} \right\}$$

$F(S)$	$\{a, b, \epsilon\}$
$F(A)$	$\{a, \epsilon\}$
$F(B)$	$\{b, \epsilon\}$

$$F(S) = F(A) \cup F(B)$$

$$\{ \epsilon \}$$

$$(4) \quad S \rightarrow \underline{aAB} \mid \underline{bBa}$$

$$A \rightarrow \epsilon$$

$$B \rightarrow \epsilon$$

	First
$F(S)$	$\{a, b\}$
$F(A)$	$\{\epsilon\}$
$F(B)$	$\{\epsilon\}$

$$\textcircled{5} \quad S \rightarrow AaAl \mid bBa$$

$$A \rightarrow d \mid e$$

$$B \rightarrow c \mid e$$

$$F(S) = \{d, a, c, b\}$$

$$F(A) = \{d, e\}$$

$$F(B) = \{c, e\}$$

$$\textcircled{6} \quad S \rightarrow aAB \mid bAaB \mid \epsilon$$

$$A \rightarrow \overset{\epsilon}{\textcircled{S}}$$

$$B \rightarrow S$$

$$F(B) = F(S) \\ \{a, b, \epsilon\}$$

$F(S)$	$\{a, b, \epsilon\}$
$F(A)$	$\{a, b, \epsilon\}$
$F(B)$	$\{a, b, \epsilon\}$

①

$$(S) \rightarrow (A)B$$

$$A \rightarrow \underline{a}b \mid b\underline{a}$$

$$B \rightarrow \underline{b}c \mid c\underline{d}$$

	First
First(S)	= First = {a, b}
First(A)	{a, b}
First(B)	{b, c}

$$T = \{+, *, (,), a\}$$

⑦

$$E \rightarrow \textcircled{T} E'$$

$$E' \rightarrow \underline{+} T E' \mid \underline{\epsilon}$$

$$\underline{T} \rightarrow \textcircled{F} T'$$

$$T' \rightarrow * F T' \mid \epsilon$$

$$F \rightarrow \textcircled{(} E \textcircled{)} \mid a$$

$$(F) = \{(, a)$$

	First
$F(E)$	$\{(, a)$
$F(E')$	$\{+, \epsilon\}$
$F(T)$	$\{(, a)$
$F(T')$	$\{*, \epsilon\}$
$F(F)$	$\{(, a)$

$$\textcircled{8} \quad S \rightarrow \underline{A} \underline{C} B \mid \underline{C} \underline{b} \mid \underline{B} \underline{a}$$

$$A \rightarrow d a \mid \underline{B} C$$

$$B \rightarrow g \mid e$$

$$C \rightarrow h \mid e$$

	First
<u>F(S)</u>	{d, g, h, e, b, a}
F(A)	{d, g, h, e}
F(B)	{g, e}
F(C)	{h, e}



Topic : Follow Set

- ① Follow (s) = { $\$$ }
- ② If Grammar is
 $A \rightarrow \alpha \underline{B} X$ $X \rightarrow \epsilon$
Follow W(B) = First (X)
- ③ If Grammar is
 $\underline{A} \rightarrow \alpha \underline{B}$
Follow (B) = Follow (A)
- ④ If Grammar is
 $A \rightarrow \underline{B} X$ $X \rightarrow \epsilon$
Follow (B) = {First (X) - ϵ } $\cup$ Follow (A)

$$\begin{aligned} \textcircled{1} \quad S &\rightarrow AB \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad S &\rightarrow AaAb / BbBa \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$



2 mins Summary



Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU